E-commerce Growth & Customer Behavior Analysis

Tools: SQL, dbt, Tableau

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Understanding Revenue Sustainability & Customer Value

Context:

This analysis evaluates the performance of a consumer electronics marketplace over a 60-day window, focusing on:

- Conversion efficiency across the funnel
- Retention and repeat purchase behavior
- Revenue concentration by brand and category
- Long-term cohort value (LTV)

Objective:

To determine whether recent revenue growth reflects sustainable customer value or short-term promotional effects.

Key findings

- **Revenue is highly concentrated,** with 90% coming from Apple and Samsung and 93% from the smartphone category, creating both growth focus and brand dependency risk.

- **The largest funnel drop occurs at the View → Add to Cart stage (3.29% CR)**, indicating significant friction before purchase intent is fully formed.

- **The November 10 revenue spike was promotion-driven**, likely reflecting pre–Black Friday demand rather than sustainable organic growth.

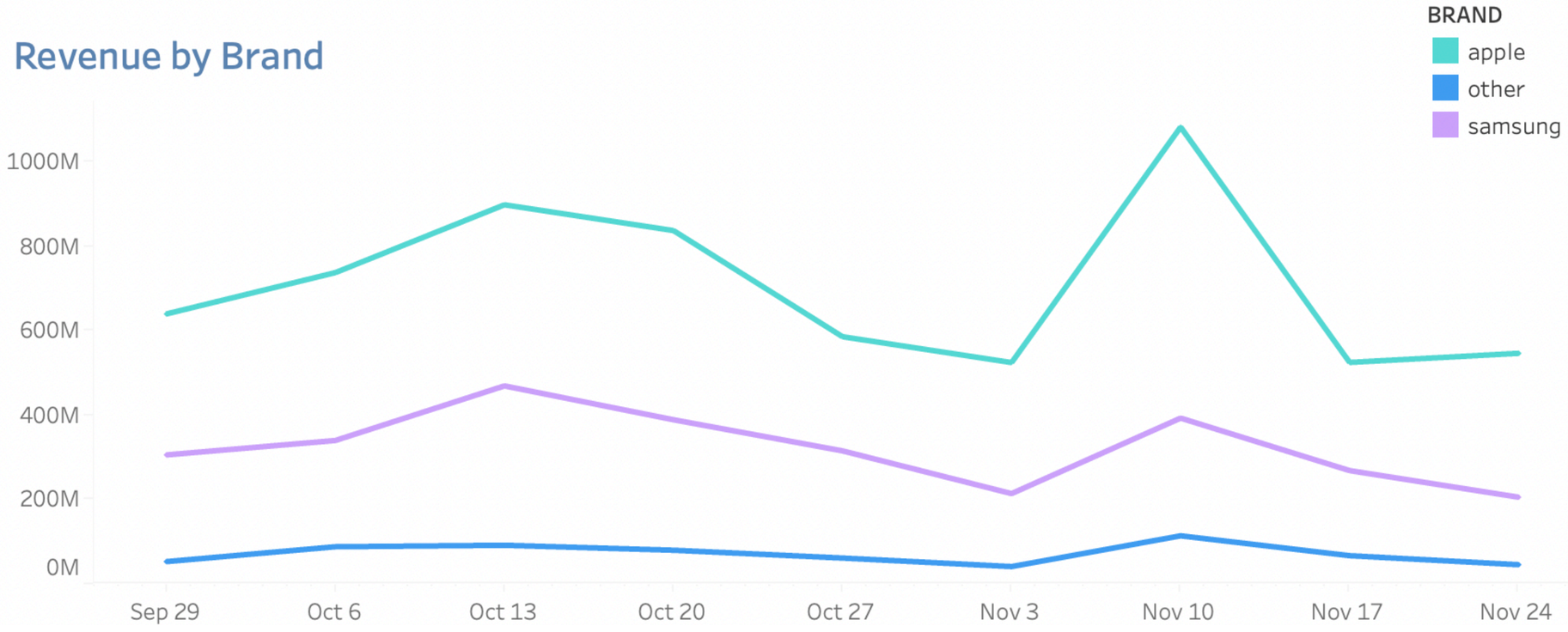
- **Repeat purchase rate is low (3.35%)**, meaning revenue is primarily driven by new customers rather than long-term retention.

- **Median time to second purchase is just 6.36 days**, suggesting a short window of opportunity for post-purchase engagement and cross-sell campaigns.

Revenue Overview

Total Revenue	ARPU	Payer Rate
$4,899,696	$7.21	1.81%

Revenue by Brand

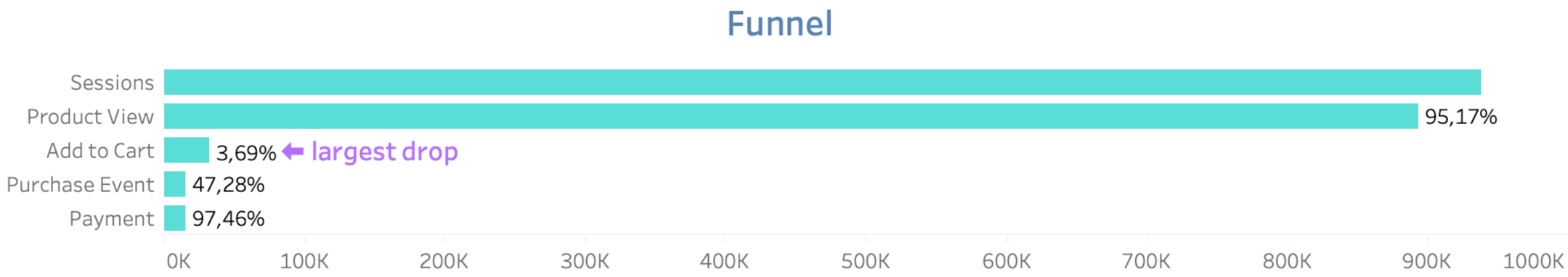


A significant revenue spike occurred in early November, likely reflecting pre–Black Friday behavior.

The business is highly sensitive to promotions. Users appear to research products in advance and convert when discounts are anticipated.

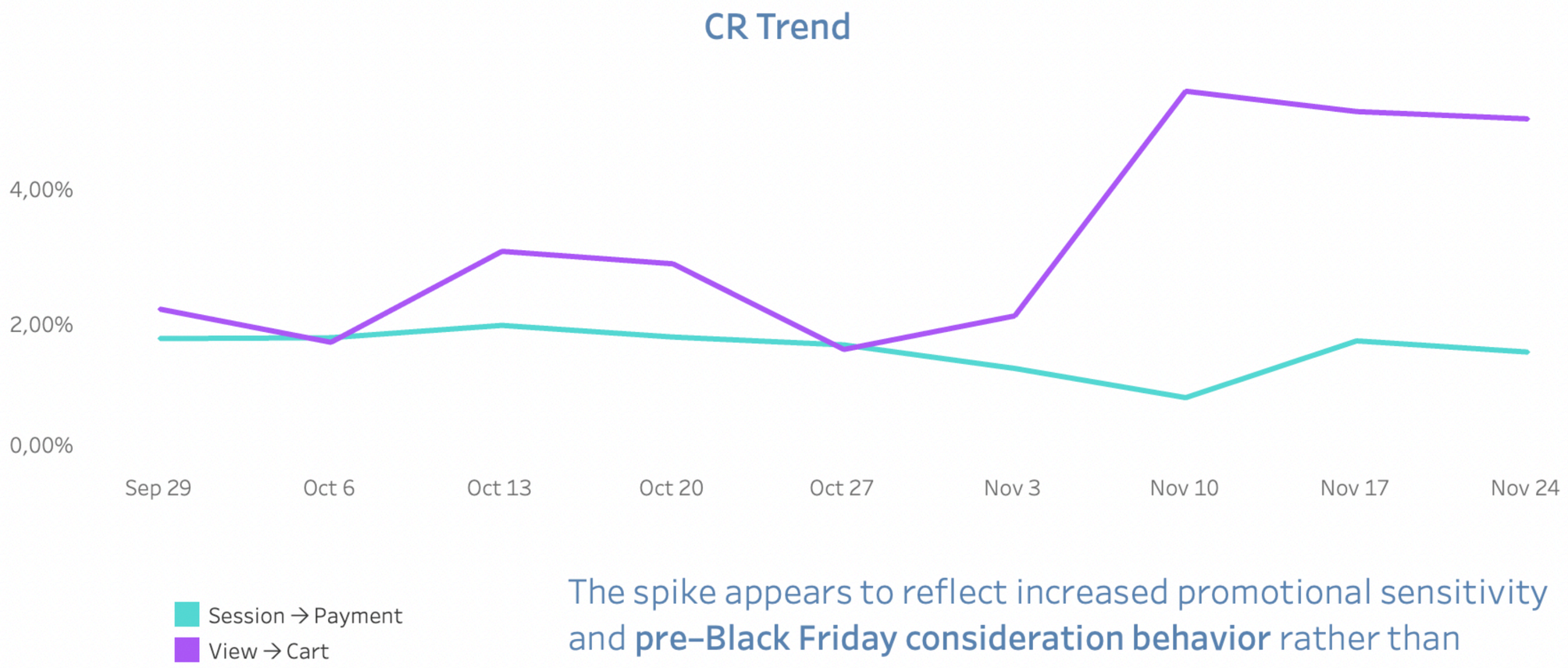
Funnel Analysis

Session → Payment CR	View → Cart CR	Payment Success Rate	Average Time to Purchase
1,63%	**3,29%**	**97,44%**	**3,34 days**



The primary bottleneck in the funnel occurs before purchase intent is formed, with the biggest drop at the Add-to-Cart stage, while checkout performance remains strong.

Conversion Trend



The spike appears to reflect increased promotional sensitivity and **pre-Black Friday consideration behavior** rather than immediate buying intent.

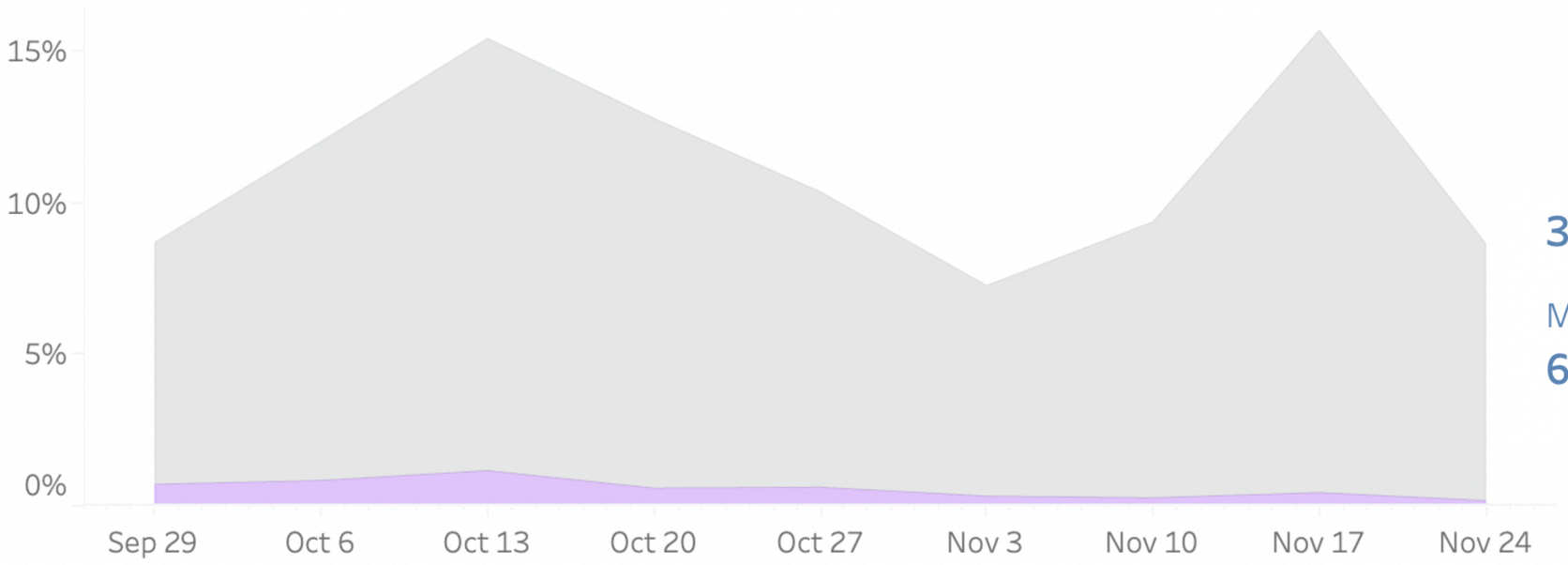
Cohort Repeat Retention Heatmap

	D1	D7	D14	D21	D30
September 30, 2019	1,30%	2,94%	3,83%	4,38%	5,20%
October 7, 2019	1,04%	2,23%	2,95%	3,63%	4,04%
October 14, 2019	1,16%	2,59%	3,47%	3,93%	4,53%
October 21, 2019	1,26%	2,13%	2,57%	3,00%	3,22%
October 28, 2019	0,65%	1,80%	2,23%	2,59%	3,09%

- Retention steadily declines for cohorts acquired in late October and November, with D30 levels dropping to around 1–2%.

- The decline in retention across later cohorts may reflect the product lifecycle effect of new smartphone launches, where customers make one planned purchase and do not return in the short term.

Revenue Composition (Single vs Repeat)

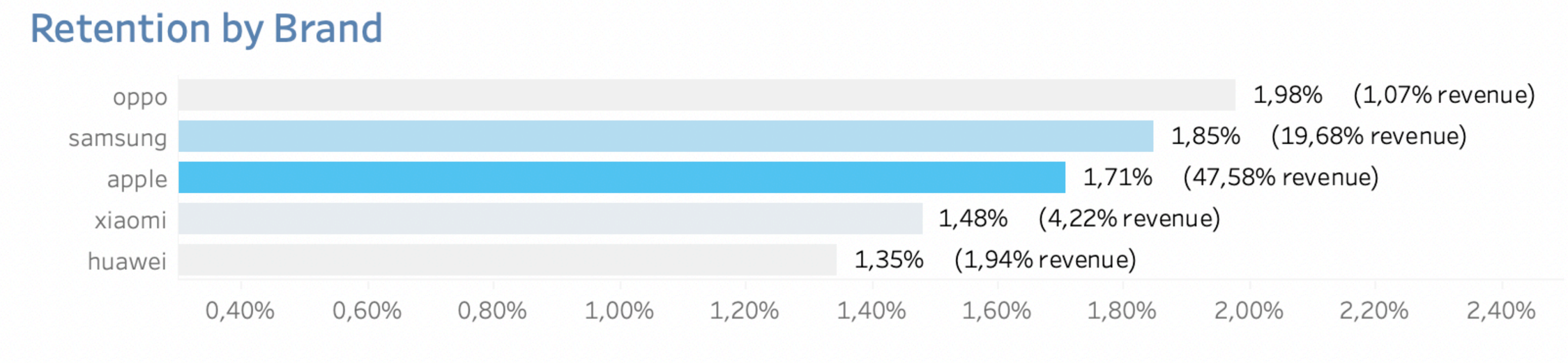


3,35% of payers made 2+ purchases

Median time to 2nd purchase:
6,36 days

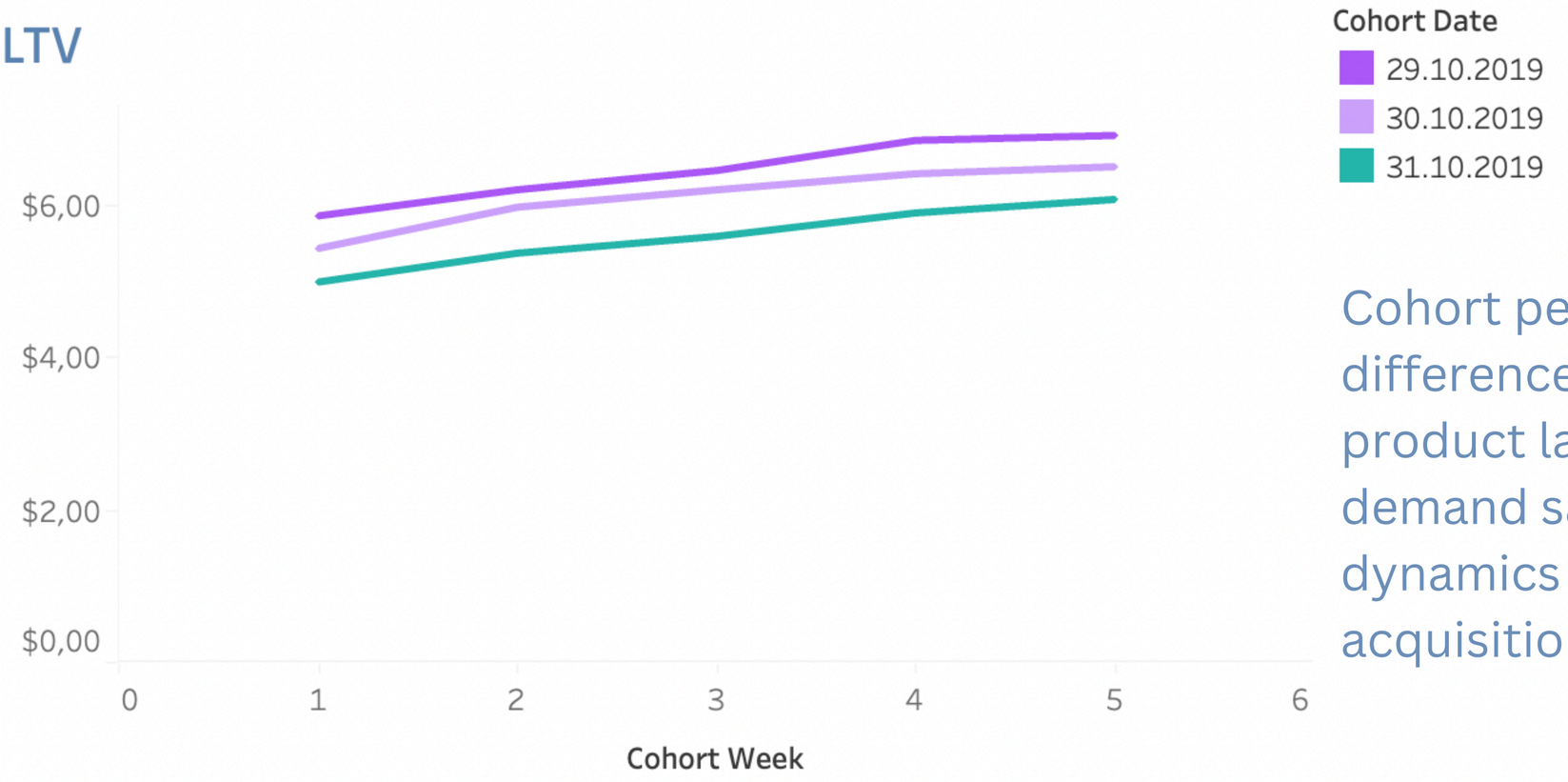
- The business relies heavily on continuous acquisition of new customers. Growth is **acquisition-driven** rather than loyalty-driven.
- Even during the revenue spike on November 17, repeat purchases did not increase proportionally.
- Retention is **not** a meaningful driver of revenue.

Retention by Brand



- Oppo has the highest retention rate (1.98%) but contributes only 1.07% of total revenue, suggesting untapped scaling potential.
- Samsung shows a balanced performance, combining solid retention (1.85%) with a meaningful revenue share (19.68%).
- Apple generates nearly half of total revenue (47.58%) but has lower retention (1.71%) than Samsung, indicating opportunity to strengthen loyalty strategies.
- Huawei and Xiaomi show both lower retention and smaller revenue shares, making them lower strategic priorities.

Cohort LTV



September 29, 2019	$6,36	$7,30	$8,04	$8,54	$8,90
October 6, 2019	$5,62	$6,31	$6,73	$7,02	$7,20
October 13, 2019	$6,49	$7,13	$7,45	$7,62	$7,75
October 20, 2019	$5,80	$6,35	$6,55	$6,76	$6,99
October 27, 2019	$5,60	$5,97	$6,18	$6,52	$6,61

Cohort performance differences may reflect product launch cycles and demand saturation dynamics rather than solely acquisition quality changes.

Strategic Opportunities

1. Improving View → Cart conversion
   - Stronger call-to-action
   - Urgency messaging
   - Price anchoring or installment visibility
2. Post-purchase cross-sell within 7 days
   - Accessory recommendations
   - Personalized follow-up campaigns
3. Promotion-driven activation
   - Targeted remarketing before major sale events
   - Early access campaigns for high-intent users
4. Revenue diversification
   - Expanding mid-tier products
   - Increasing share of non-smartphone categories